Quantium Virtual Internship - Retail Strategy and Analytics - Task 1

Bibliotecas e datasets utilizados:

```r
library(data.table)
```

```
## Warning: pacote 'data.table' foi compilado no R versão 4.4.3
```

```r
library(readxl)
```

```
## Warning: pacote 'readxl' foi compilado no R versão 4.4.2
```

```r
library(tidyverse)
```

```
## Warning: pacote 'tidyverse' foi compilado no R versão 4.4.3

## Warning: pacote 'ggplot2' foi compilado no R versão 4.4.3

## Warning: pacote 'tibble' foi compilado no R versão 4.4.2

## Warning: pacote 'tidyr' foi compilado no R versão 4.4.3

## Warning: pacote 'readr' foi compilado no R versão 4.4.3

## Warning: pacote 'purrr' foi compilado no R versão 4.4.3

## Warning: pacote 'dplyr' foi compilado no R versão 4.4.3

## Warning: pacote 'stringr' foi compilado no R versão 4.4.3

## Warning: pacote 'forcats' foi compilado no R versão 4.4.3

## Warning: pacote 'lubridate' foi compilado no R versão 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.2     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.4
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
```

```
## x dplyr::between()     masks data.table::between()
## x dplyr::filter()      masks stats::filter()
## x dplyr::first()       masks data.table::first()
## x lubridate::hour()    masks data.table::hour()
## x lubridate::isoweek() masks data.table::isoweek()
## x dplyr::lag()         masks stats::lag()
## x dplyr::last()        masks data.table::last()
## x lubridate::mday()    masks data.table::mday()
## x lubridate::minute()  masks data.table::minute()
## x lubridate::month()   masks data.table::month()
## x lubridate::quarter() masks data.table::quarter()
## x lubridate::second()  masks data.table::second()
## x purrr::transpose()   masks data.table::transpose()
## x lubridate::wday()    masks data.table::wday()
## x lubridate::week()    masks data.table::week()
## x lubridate::yday()    masks data.table::yday()
## x lubridate::year()    masks data.table::year()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(tinytex)
```

```
## Warning: pacote 'tinytex' foi compilado no R versão 4.4.3
```

```r
library(stringr)

filePath <- "C:/Users/renat/Documents/quantium/quantium_chips/"
transactionData <- read_xlsx(paste0(filePath,"QVI_transaction_data.xlsx"))
customerData <- fread(paste0(filePath,"QVI_purchase_behaviour.csv"))
transactionData <- as.data.table(transactionData)
customerData <- as.data.table(customerData)
```

Primeiramente queremos entender os dados que iremos trabalhar. Vamos explorar os dois datasets fornecidos e fazer a limpeza de dados se necessário.

Customer Data

```r
str(customerData)
```

```
## Classes 'data.table' and 'data.frame':   72637 obs. of  3 variables:
##  $ LYLTY_CARD_NBR  : int  1000 1002 1003 1004 1005 1007 1009 1010 1011 1012 ...
##  $ LIFESTAGE       : chr  "YOUNG SINGLES/COUPLES" "YOUNG SINGLES/COUPLES" "YOUNG FAMILIES" "OLDER SI
##  $ PREMIUM_CUSTOMER: chr  "Premium" "Mainstream" "Budget" "Mainstream" ...
##  - attr(*, ".internal.selfref")=<externalptr>
```

```r
head(customerData)
```

```
##    LYLTY_CARD_NBR              LIFESTAGE PREMIUM_CUSTOMER
##             <int>                 <char>           <char>
## 1:           1000  YOUNG SINGLES/COUPLES          Premium
## 2:           1002  YOUNG SINGLES/COUPLES       Mainstream
## 3:           1003         YOUNG FAMILIES           Budget
```

```
## 4:           1004  OLDER SINGLES/COUPLES       Mainstream
## 5:           1005 MIDAGE SINGLES/COUPLES       Mainstream
## 6:           1007  YOUNG SINGLES/COUPLES           Budget
```

Neste dataset temos os dados dos clientes, onde cada clente tem um número de cartão de lealdade, e categorias como estágio de vida e cliente premium.

Transaction data

```r
str(transactionData)
```

```
## Classes 'data.table' and 'data.frame':   264836 obs. of  8 variables:
##  $ DATE          : num  43390 43599 43605 43329 43330 ...
##  $ STORE_NBR     : num  1 1 1 2 2 4 4 4 5 7 ...
##  $ LYLTY_CARD_NBR: num  1000 1307 1343 2373 2426 ...
##  $ TXN_ID        : num  1 348 383 974 1038 ...
##  $ PROD_NBR      : num  5 66 61 69 108 57 16 24 42 52 ...
##  $ PROD_NAME     : chr  "Natural Chip        Compny SeaSalt175g" "CCs Nacho Cheese    175g" "Smiths C
##  $ PROD_QTY      : num  2 3 2 5 3 1 1 1 1 2 ...
##  $ TOT_SALES     : num  6 6.3 2.9 15 13.8 5.1 5.7 3.6 3.9 7.2 ...
##  - attr(*, ".internal.selfref")=<externalptr>
```

```r
head(transactionData)
```

```
##     DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR
##    <num>     <num>          <num>  <num>    <num>
## 1: 43390         1           1000      1        5
## 2: 43599         1           1307    348       66
## 3: 43605         1           1343    383       61
## 4: 43329         2           2373    974       69
## 5: 43330         2           2426   1038      108
## 6: 43604         4           4074   2982       57
##                                  PROD_NAME PROD_QTY TOT_SALES
##                                     <char>    <num>     <num>
## 1:   Natural Chip        Compny SeaSalt175g        2       6.0
## 2:                 CCs Nacho Cheese    175g        3       6.3
## 3:   Smiths Crinkle Cut  Chips Chicken 170g        2       2.9
## 4:   Smiths Chip Thinly  S/Cream&Onion 175g        5      15.0
## 5: Kettle Tortilla ChpsHny&Jlpno Chili 150g        3      13.8
## 6: Old El Paso Salsa   Dip Tomato Mild 300g        1       5.1
```

Neste dataset temos dados de transações feitas em um ano.

Como a coluna DATE está em formato numérico, precisamos converter a coluna para formato em data.

```r
transactionData$DATE <- as.Date(transactionData$DATE, origin = "1899-12-30")
```

Vamos verficar se todos os produtos comprados são realmente chips examinando a coluna PROD_NAME.

```r
table(transactionData$PROD_NAME)
```

```
##
##                        Burger Rings 220g
##                                     1564
##                CCs Nacho Cheese    175g
##                                     1498
##                        CCs Original 175g
##                                     1514
##                CCs Tasty Cheese    175g
##                                     1539
##           Cheetos Chs & Bacon Balls 190g
##                                     1479
##                       Cheetos Puffs 165g
##                                     1448
##                    Cheezels Cheese 330g
##                                     3149
##                Cheezels Cheese Box 125g
##                                     1454
##           Cobs Popd Sea Salt  Chips 110g
##                                     3265
##   Cobs Popd Sour Crm  &Chives Chips 110g
##                                     3159
## Cobs Popd Swt/Chlli &Sr/Cream Chips 110g
##                                     3269
##         Dorito Corn Chp     Supreme 380g
##                                     3185
##         Doritos Cheese      Supreme 330g
##                                     3052
##  Doritos Corn Chip Mexican Jalapeno 150g
##                                     3204
##  Doritos Corn Chip Southern Chicken 150g
##                                     3172
##  Doritos Corn Chips  Cheese Supreme 170g
##                                     3217
##    Doritos Corn Chips  Nacho Cheese 170g
##                                     3160
##        Doritos Corn Chips  Original 170g
##                                     3121
##                Doritos Mexicana    170g
##                                     3115
##          Doritos Salsa       Medium 300g
##                                     1449
##                Doritos Salsa Mild  300g
##                                     1472
##           French Fries Potato Chips 175g
##                                     1418
##    Grain Waves         Sweet Chilli 210g
##                                     3167
##    Grain Waves Sour    Cream&Chives 210G
##                                     3105
##    GrnWves Plus Btroot & Chilli Jam 180g
##                                     1468
##  Infuzions BBQ Rib   Prawn Crackers 110g
##                                     3174
##  Infuzions Mango     Chutny Papadums 70g
```

```
##                                     1507
## Infuzions SourCream&Herbs Veg Strws 110g
##                                     3134
## Infuzions Thai SweetChili PotatoMix 110g
##                                     3242
##   Infzns Crn Crnchers Tangy Gcamole 110g
##                                     3144
##             Kettle 135g Swt Pot Sea Salt
##                                     3257
##                       Kettle Chilli 175g
##                                     3038
##         Kettle Honey Soy    Chicken 175g
##                                     3148
##   Kettle Mozzarella   Basil & Pesto 175g
##                                     3304
##                     Kettle Original 175g
##                                     3159
##     Kettle Sea Salt     And Vinegar 175g
##                                     3173
##       Kettle Sensations   BBQ&Maple 150g
##                                     3083
## Kettle Sensations   Camembert & Fig 150g
##                                     3219
##    Kettle Sensations   Siracha Lime 150g
##                                     3127
##  Kettle Sweet Chilli And Sour Cream 175g
##                                     3200
##  Kettle Tortilla ChpsBtroot&Ricotta 150g
##                                     3146
##     Kettle Tortilla ChpsFeta&Garlic 150g
##                                     3138
## Kettle Tortilla ChpsHny&Jlpno Chili 150g
##                                     3296
##   Natural Chip        Compny SeaSalt175g
##                                     1468
##  Natural Chip Co     Tmato Hrb&Spce 175g
##                                     1572
##   Natural ChipCo      Hony Soy Chckn175g
##                                     1460
##   Natural ChipCo Sea  Salt & Vinegr 175g
##                                     1550
##   NCC Sour Cream &    Garden Chives 175g
##                                     1419
## Old El Paso Salsa   Dip Chnky Tom Ht300g
##                                     3125
##  Old El Paso Salsa   Dip Tomato Med 300g
##                                     3114
## Old El Paso Salsa   Dip Tomato Mild 300g
##                                     3085
##                 Pringles Barbeque   134g
##                                     3210
##      Pringles Chicken    Salt Crips 134g
##                                     3104
##         Pringles Mystery    Flavour 134g
```

```
##                                    3114
##          Pringles Original   Crisps 134g
##                                    3157
##                 Pringles Slt Vingar 134g
##                                    3095
##           Pringles SourCream  Onion 134g
##                                    3162
##         Pringles Sthrn FriedChicken 134g
##                                    3083
##             Pringles Sweet&Spcy BBQ 134g
##                                    3177
##    Red Rock Deli Chikn&Garlic Aioli 150g
##                                    1434
##  Red Rock Deli Sp    Salt & Truffle 150G
##                                    1498
## Red Rock Deli SR    Salsa & Mzzrlla 150g
##                                    1458
##     Red Rock Deli Thai  Chilli&Lime 150g
##                                    1495
##         RRD Chilli&         Coconut 150g
##                                    1506
##         RRD Honey Soy       Chicken 165g
##                                    1513
##                 RRD Lime & Pepper   165g
##                                    1473
##                 RRD Pc Sea Salt     165g
##                                    1431
##                 RRD Salt & Vinegar  165g
##                                    1474
##      RRD SR Slow Rst     Pork Belly 150g
##                                    1526
##     RRD Steak &         Chimuchurri 150g
##                                    1455
##      RRD Sweet Chilli &  Sour Cream 165g
##                                    1516
##       Smith Crinkle Cut   Bolognese 150g
##                                    1451
##    Smith Crinkle Cut   Mac N Cheese 150g
##                                    1512
##    Smiths Chip Thinly  Cut Original 175g
##                                    1614
##   Smiths Chip Thinly  CutSalt/Vinegr175g
##                                    1440
##   Smiths Chip Thinly  S/Cream&Onion 175g
##                                    1473
##        Smiths Crinkle      Original 330g
##                                    3142
## Smiths Crinkle Chips Salt & Vinegar 330g
##                                    3197
##  Smiths Crinkle Cut  Chips Barbecue 170g
##                                    1489
##   Smiths Crinkle Cut  Chips Chicken 170g
##                                    1484
##  Smiths Crinkle Cut  Chips Chs&Onion170g
```

```
##                                      1481
##  Smiths Crinkle Cut  Chips Original 170g
##                                      1461
## Smiths Crinkle Cut  French OnionDip 150g
##                                      1438
##  Smiths Crinkle Cut  Salt & Vinegar 170g
##                                      1455
##      Smiths Crinkle Cut  Snag&Sauce 150g
##                                      1503
##    Smiths Crinkle Cut  Tomato Salsa 150g
##                                      1470
##   Smiths Crnkle Chip  Orgnl Big Bag 380g
##                                      3233
## Smiths Thinly       Swt Chli&S/Cream175G
##                                      1461
##   Smiths Thinly Cut   Roast Chicken 175g
##                                      1519
##     Snbts Whlgrn Crisps Cheddr&Mstrd 90g
##                                      1576
## Sunbites Whlegrn    Crisps Frch/Onin 90g
##                                      1432
##   Thins Chips         Originl saltd 175g
##                                      1441
##           Thins Chips Light&  Tangy 175g
##                                      3188
##         Thins Chips Salt &  Vinegar 175g
##                                      3103
##         Thins Chips Seasonedchicken 175g
##                                      3114
##     Thins Potato Chips  Hot & Spicy 175g
##                                      3229
##          Tostitos Lightly    Salted 175g
##                                      3074
##        Tostitos Smoked     Chipotle 175g
##                                      3145
##            Tostitos Splash Of  Lime 175g
##                                      3252
##                 Twisties Cheese     270g
##                                      3115
##          Twisties Cheese     Burger 250g
##                                      3169
##                     Twisties Chicken270g
##                                      3170
##   Tyrrells Crisps     Ched & Chives 165g
##                                      3268
##  Tyrrells Crisps     Lightly Salted 165g
##                                      3174
##           Woolworths Cheese   Rings 190g
##                                      1516
##           Woolworths Medium   Salsa 300g
##                                      1430
##           Woolworths Mild     Salsa 300g
##                                      1491
##         WW Crinkle Cut      Chicken 175g
```

```
##                                      1467
##        WW Crinkle Cut      Original 175g
##                                      1410
##        WW D/Style Chip     Sea Salt 200g
##                                      1469
##           WW Original Corn    Chips 200g
##                                      1495
##           WW Original Stacked Chips 160g
##                                      1487
##   WW Sour Cream &OnionStacked Chips 160g
##                                      1483
##      WW Supreme Cheese   Corn Chips 200g
##                                      1509
```

```
productWords <- data.table(unlist(strsplit(unique(transactionData$PROD_NAME)," ")))
setnames(productWords,'words')
```

Removeremos dígitos e carcteres especiais.

```
productWords <- productWords[grepl("^[a-zA-Z]+$", words)]
```

Tabela de frequencias das palavras.

```
freq_palavras <- productWords[, .N, by = words][order(-N)]
```

Queremos remover produtos que não são chips, como salsa.

```
transactionData[, SALSA := grepl("salsa", tolower(PROD_NAME))]
transactionData <- transactionData[SALSA == FALSE, ][, SALSA := NULL]
```

Sumario da tabela transactionData. Queremos verificar possíveis valores nulos e outliers.

```
summary(transactionData)
```

```
##       DATE              STORE_NBR     LYLTY_CARD_NBR        TXN_ID
##  Min.   :2018-07-01   Min.   :  1.0   Min.   :   1000   Min.   :      1
##  1st Qu.:2018-09-30   1st Qu.: 70.0   1st Qu.:  70015   1st Qu.:  67569
##  Median :2018-12-30   Median :130.0   Median : 130367   Median : 135183
##  Mean   :2018-12-30   Mean   :135.1   Mean   : 135531   Mean   : 135131
##  3rd Qu.:2019-03-31   3rd Qu.:203.0   3rd Qu.: 203084   3rd Qu.: 202654
##  Max.   :2019-06-30   Max.   :272.0   Max.   :2373711   Max.   :2415841
##     PROD_NBR       PROD_NAME            PROD_QTY         TOT_SALES
##  Min.   :  1.00   Length:246742      Min.   :  1.000   Min.   :  1.700
##  1st Qu.: 26.00   Class :character   1st Qu.:  2.000   1st Qu.:  5.800
##  Median : 53.00   Mode  :character   Median :  2.000   Median :  7.400
##  Mean   : 56.35                      Mean   :  1.908   Mean   :  7.321
##  3rd Qu.: 87.00                      3rd Qu.:  2.000   3rd Qu.:  8.800
##  Max.   :114.00                      Max.   :200.000   Max.   :650.000
```

Não temos valores nulos, porem, o PROD_QTY aparenta ter outliers(max 200 pacotes vendidos). Vamos verificar:

```r
transactionData[PROD_QTY == 200]
```

```
##          DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR
##        <Date>     <num>          <num>  <num>    <num>
## 1: 2018-08-19       226         226000 226201        4
## 2: 2019-05-20       226         226000 226210        4
##                           PROD_NAME PROD_QTY TOT_SALES
##                              <char>    <num>     <num>
## 1: Dorito Corn Chp     Supreme 380g      200       650
## 2: Dorito Corn Chp     Supreme 380g      200       650
```

Encontramos duas transações de 200 pacotes, de um mesmo cliente(LYLTY_CARD_NBR = 226000). Vamos Verificar se este cliente fez mais transações além dessas duas.

```r
transactionData[LYLTY_CARD_NBR == 226000]
```

```
##          DATE STORE_NBR LYLTY_CARD_NBR TXN_ID PROD_NBR
##        <Date>     <num>          <num>  <num>    <num>
## 1: 2018-08-19       226         226000 226201        4
## 2: 2019-05-20       226         226000 226210        4
##                           PROD_NAME PROD_QTY TOT_SALES
##                              <char>    <num>     <num>
## 1: Dorito Corn Chp     Supreme 380g      200       650
## 2: Dorito Corn Chp     Supreme 380g      200       650
```

Parece que este cliente tem somente duas transações durante o ano e não é um cliente ordinário. Vamos remover este cliente para futuras análises.

```r
transactionData <- transactionData[LYLTY_CARD_NBR != 226000]
```

```r
summary(transactionData)
```

```
##       DATE              STORE_NBR     LYLTY_CARD_NBR        TXN_ID
##  Min.   :2018-07-01   Min.   :  1.0   Min.   :   1000   Min.   :      1
##  1st Qu.:2018-09-30   1st Qu.: 70.0   1st Qu.:  70015   1st Qu.:  67569
##  Median :2018-12-30   Median :130.0   Median : 130367   Median : 135182
##  Mean   :2018-12-30   Mean   :135.1   Mean   : 135530   Mean   : 135130
##  3rd Qu.:2019-03-31   3rd Qu.:203.0   3rd Qu.: 203083   3rd Qu.: 202652
##  Max.   :2019-06-30   Max.   :272.0   Max.   :2373711   Max.   :2415841
##     PROD_NBR       PROD_NAME            PROD_QTY       TOT_SALES
##  Min.   :  1.00   Length:246740      Min.   :1.000   Min.   : 1.700
##  1st Qu.: 26.00   Class :character   1st Qu.:2.000   1st Qu.: 5.800
##  Median : 53.00   Mode  :character   Median :2.000   Median : 7.400
##  Mean   : 56.35                      Mean   :1.906   Mean   : 7.316
##  3rd Qu.: 87.00                      3rd Qu.:2.000   3rd Qu.: 8.800
##  Max.   :114.00                      Max.   :5.000   Max.   :29.500
```

Contaremos o número de transações por dia para verificar se temos prolemas nos dados.

```r
transacoes_por_dia <- transactionData[, .N, by = DATE]
summary(transacoes_por_dia)
```

```
##       DATE                  N
##  Min.   :2018-07-01   Min.   :607.0
##  1st Qu.:2018-09-29   1st Qu.:658.0
##  Median :2018-12-30   Median :674.0
##  Mean   :2018-12-30   Mean   :677.9
##  3rd Qu.:2019-03-31   3rd Qu.:694.2
##  Max.   :2019-06-30   Max.   :865.0
```

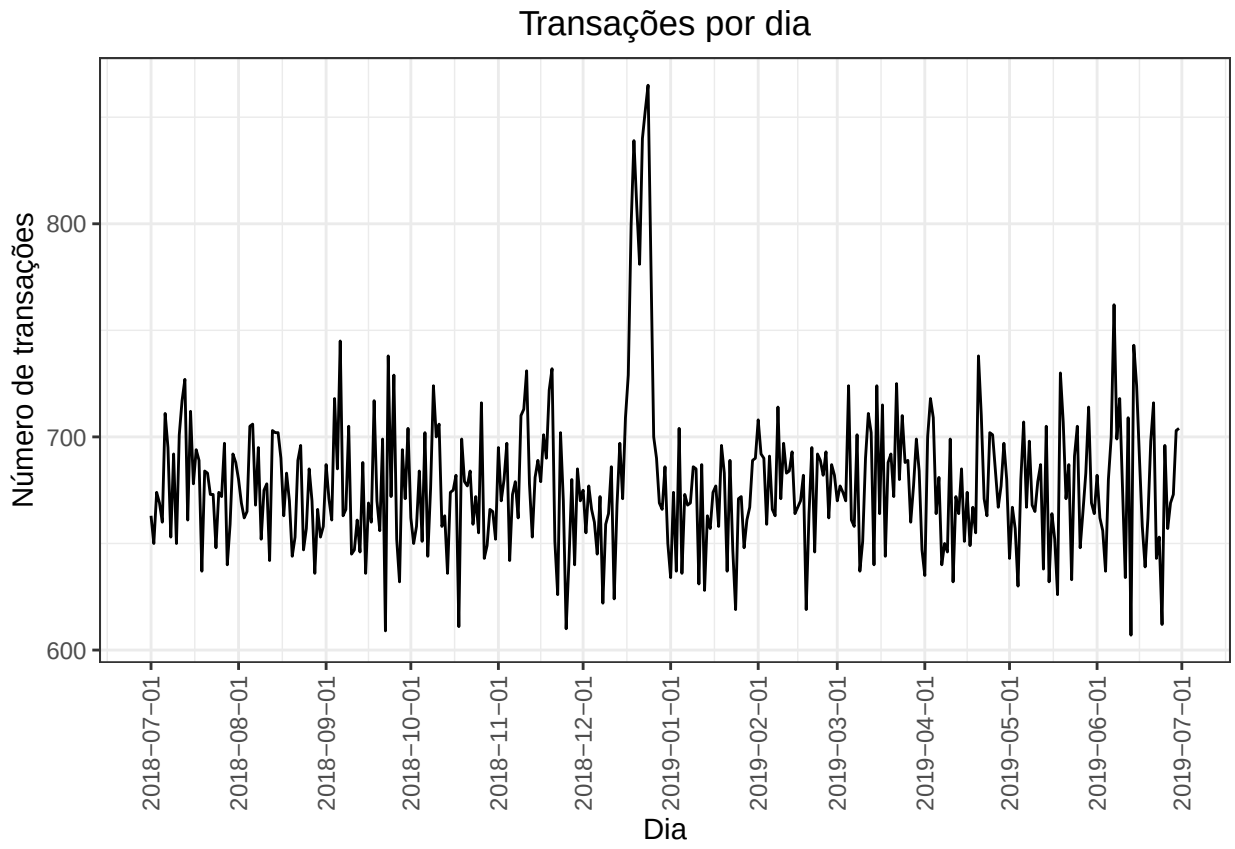
Percebemos que os dados são coletados de 01/07/2018 até 30/06/2019, porém temos 364 observações em transacoes_por_dia. Possivelmente temos uma data faltando.

Vamos criar uma tabela com todas as datas de 01/07/2018 até 30/06/2019 e comparar com o nossos dados para encontrar a data faltando.

```r
todas_datas <- data.table(
  DATE = seq(as.Date("2018-07-01"), as.Date("2019-06-30"), by = "day")
)

calendario_completo <- merge(todas_datas, transacoes_por_dia, by = "DATE", all.x = TRUE)
```

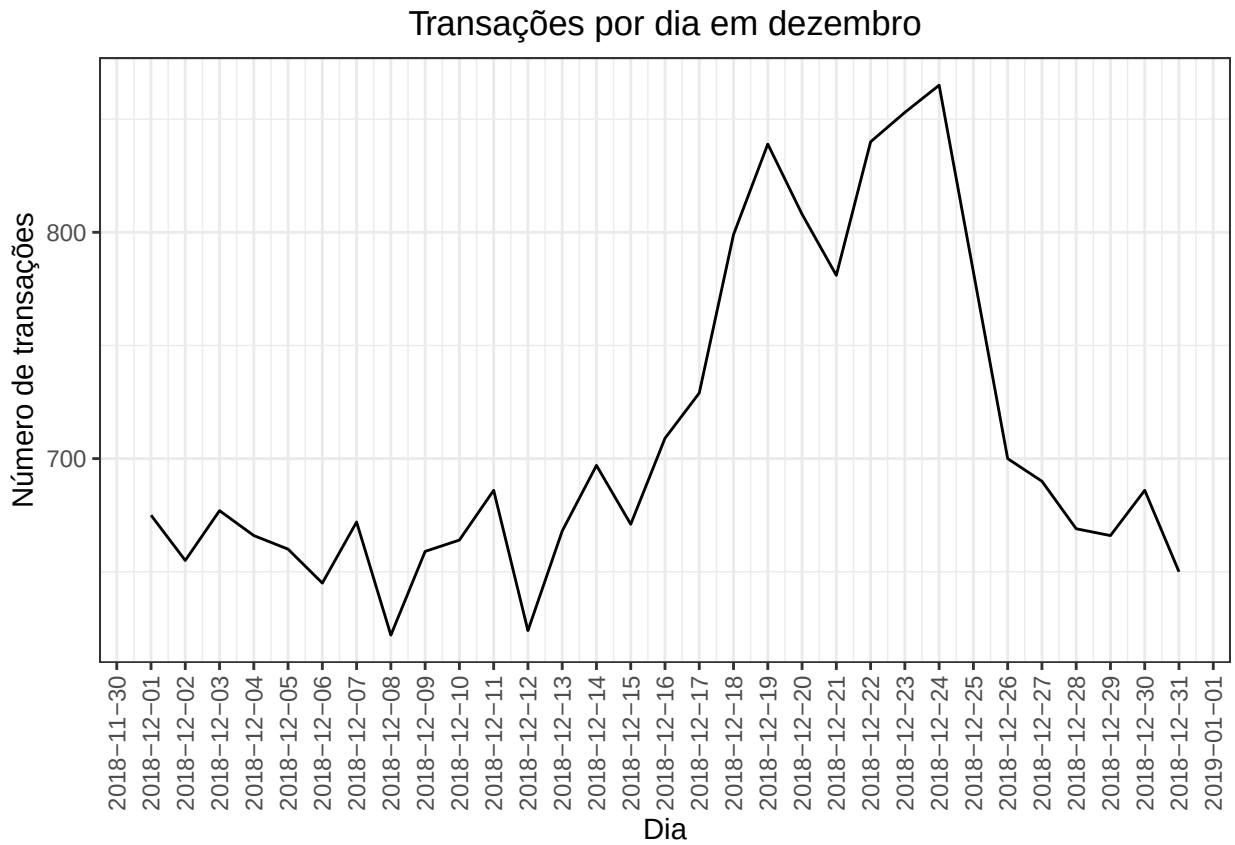
Encontramos a data faltante, dia 25/12/2018.

```r
theme_set(theme_bw())
theme_update(plot.title = element_text(hjust = 0.5))

ggplot(transacoes_por_dia, aes(x = DATE, y = N)) +
 geom_line() +
 labs(x = "Dia", y = "Número de transações", title = "Transações por dia") +
 scale_x_date(breaks = "1 month") +
 theme(axis.text.x = element_text(angle = 90, vjust = 0.5))
```



Vamos dar um zoom no mês de dezembro, onde há um crescimento nas vendas.

```
mes_dezembro <- transacoes_por_dia[month(DATE) == 12]

theme_set(theme_bw())
theme_update(plot.title = element_text(hjust = 0.5))

ggplot(mes_dezembro, aes(x = DATE, y = N)) +
 geom_line() +
 labs(x = "Dia", y = "Número de transações", title = "Transações por dia em dezembro") +
 scale_x_date(breaks = "1 day") +
 theme(axis.text.x = element_text(angle = 90, vjust = 0.5))
```



Percebemos que há um aumento de venda de chips nos dias anteriores ao Natal, porém no dia exato do Natal(25/12/2018) não há vendas, pois as lojas fecham no Natal.

Agora que os dados não tem mais outliers, vamos criar novas características como tamanho do pacote ou marca dos chips pela coluna PROD_NAME.

```
transactionData[, PACK_SIZE := parse_number(PROD_NAME)]

transactionData[, .N, PACK_SIZE][order(PACK_SIZE)]
```

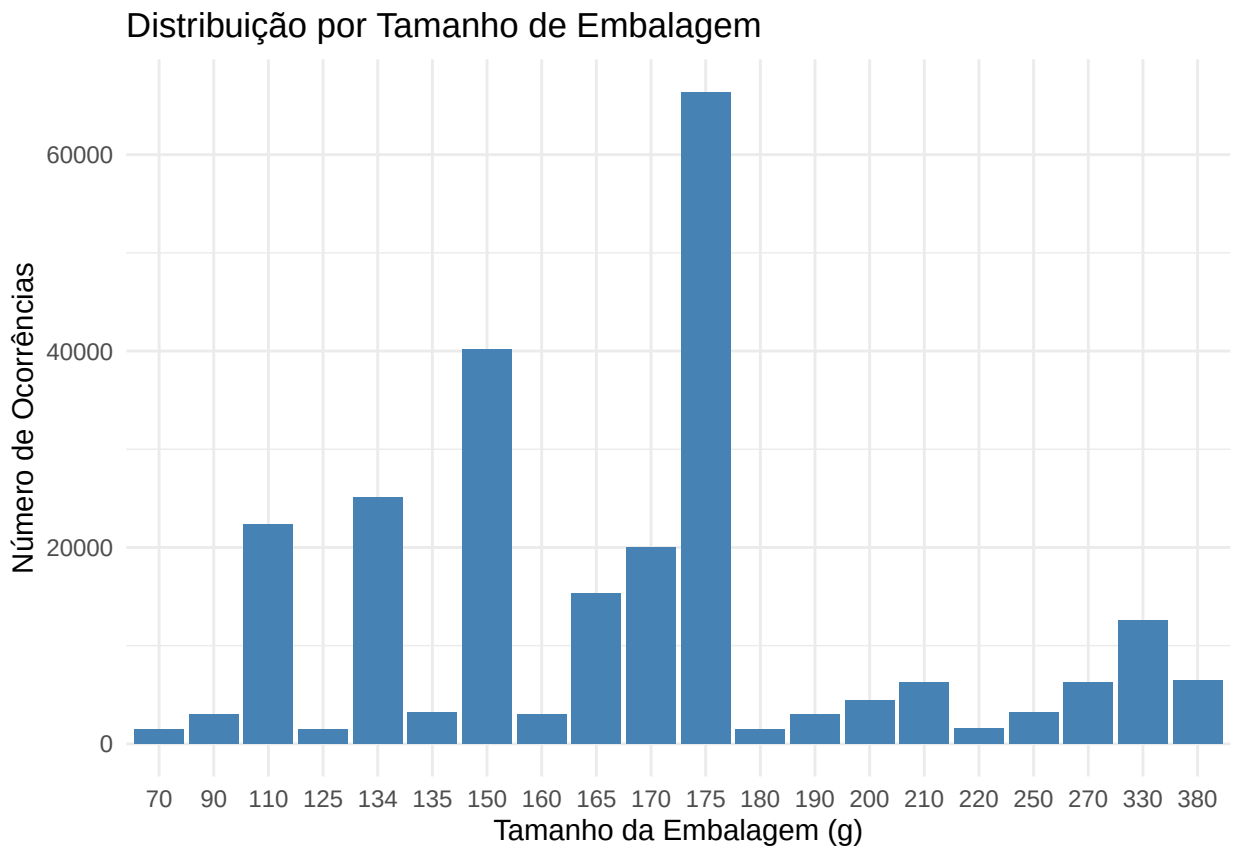
```
##     PACK_SIZE     N
##         <num> <int>
##  1:        70  1507
##  2:        90  3008
##  3:       110 22387
##  4:       125  1454
##  5:       134 25102
##  6:       135  3257
##  7:       150 40203
##  8:       160  2970
##  9:       165 15297
## 10:       170 19983
## 11:       175 66390
## 12:       180  1468
## 13:       190  2995
## 14:       200  4473
```

```
## 15:       210  6272
## 16:       220  1564
## 17:       250  3169
## 18:       270  6285
## 19:       330 12540
## 20:       380  6416
##     PACK_SIZE     N
```

Temos então pacotes de 70g até 380g, o que parece sensato.

Vamos fazer um histograma de PACK_SIZE

```
packsizeN <- transactionData[, .N, PACK_SIZE]

ggplot(packsizeN, aes(x = factor(PACK_SIZE), y = N)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Distribuição por Tamanho de Embalagem",
       x = "Tamanho da Embalagem (g)",
       y = "Número de Ocorrências") +
  theme_minimal()
```



Vamos adicionar também na tabela transactionData a característica marca do produto BRAND.

```
transactionData[, BRAND := word(PROD_NAME, 1)]

transactionData[, .N, BRAND][order(BRAND)]
```

```
##          BRAND     N
##         <char> <int>
##  1:     Burger  1564
##  2:        CCs  4551
##  3:    Cheetos  2927
##  4:   Cheezels  4603
##  5:       Cobs  9693
##  6:     Dorito  3183
##  7:    Doritos 22041
##  8:     French  1418
##  9:      Grain  6272
## 10:    GrnWves  1468
## 11:  Infuzions 11057
## 12:     Infzns  3144
## 13:     Kettle 41288
## 14:        NCC  1419
## 15:    Natural  6050
## 16:   Pringles 25102
## 17:        RRD 11894
## 18:        Red  4427
## 19:      Smith  2963
## 20:     Smiths 27390
## 21:      Snbts  1576
## 22:   Sunbites  1432
## 23:      Thins 14075
## 24:   Tostitos  9471
## 25:   Twisties  9454
## 26:   Tyrrells  6442
## 27:         WW 10320
## 28: Woolworths  1516
##          BRAND     N
```

Obtemos 28 tipos de marca de chips, o que parece sensato. Temos algumas marcas que parecem ser o mesmo, então vamos combinar estes juntos.(Red/RRD, Dorito/Doritos, Grain/GrnWves, Infuzions/Infzns, Smith/Smiths, Snbts/Sunbites, WW/Woolworths)

```r
transactionData[BRAND == "Red", BRAND := "RRD"]
transactionData[BRAND == "Dorito", BRAND := "Doritos"]
transactionData[BRAND == "GrnWves", BRAND := "Grain"]
transactionData[BRAND == "Infzns", BRAND := "Infuzions"]
transactionData[BRAND == "Smith", BRAND := "Smiths"]
transactionData[BRAND == "Snbts", BRAND := "Sunbites"]
transactionData[BRAND == "WW", BRAND := "Woolworths"]
```

Temos então 21 tipos de marca de chips.

Vamos agora observar o dataset dos clientes, customerData.

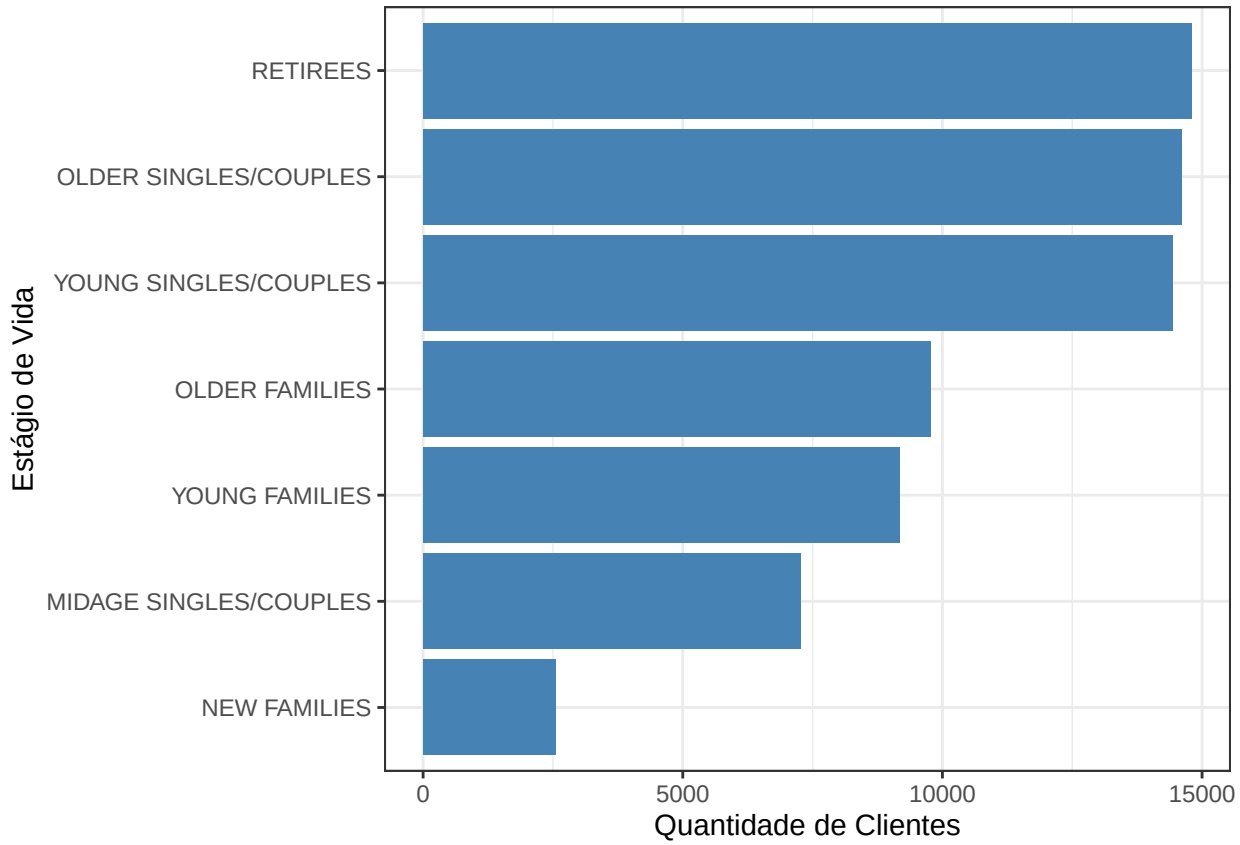
```r
summary(customerData)
```

```
##  LYLTY_CARD_NBR     LIFESTAGE         PREMIUM_CUSTOMER
##  Min.   :   1000   Length:72637       Length:72637
##  1st Qu.:  66202   Class :character   Class :character
##  Median : 134040   Mode  :character   Mode  :character
```

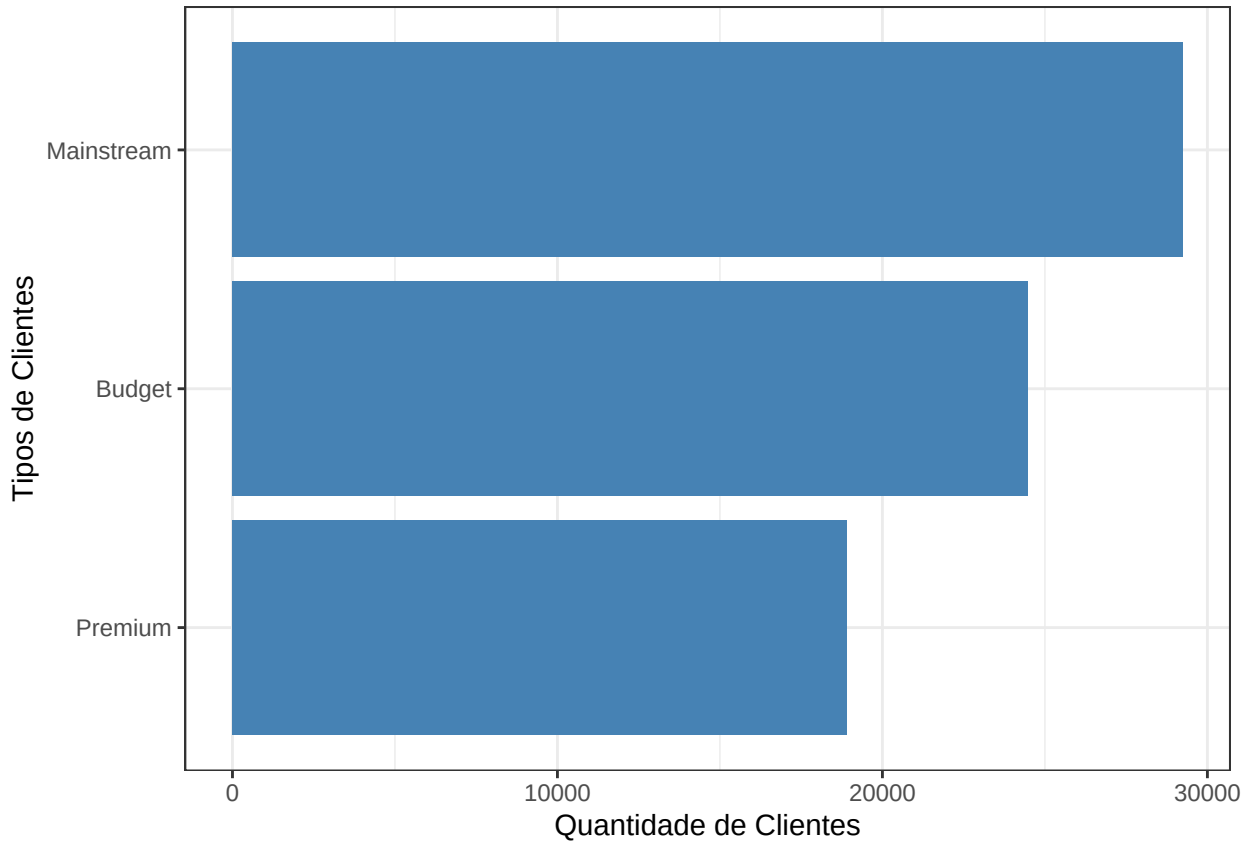
```
##  Mean   : 136186
##  3rd Qu.: 203375
##  Max.   :2373711
```

Vamos fazer o histograma de LIFESTAGE e PREMIUM_CUSTOMER

```
LifeN <- customerData[, .N, LIFESTAGE]

ggplot(LifeN, aes(x = N, y = reorder(LIFESTAGE, N))) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(x = "Quantidade de Clientes", y = "Estágio de Vida")
```




```
PremCusN <- customerData[, .N, PREMIUM_CUSTOMER]

ggplot(PremCusN, aes(x = N, y = reorder(PREMIUM_CUSTOMER, N))) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(x = "Quantidade de Clientes", y = "Tipos de Clientes")
```



Vamos criar uma tabela juntando os dados de clientes com transações

```
data <- merge(transactionData, customerData, all.x = TRUE)
```

Como o número de linhas da tabela data é o mesmo da tabela transactioData, podemos confirmar que nenhuma duplicata foi criada.

Vamos confirmar se não teve nenhum dado faltante(NULLs).

```
summary(data)
```

```
##  LYLTY_CARD_NBR         DATE               STORE_NBR         TXN_ID
##  Min.   :   1000   Min.   :2018-07-01   Min.   :  1.0   Min.   :      1
##  1st Qu.:  70015   1st Qu.:2018-09-30   1st Qu.: 70.0   1st Qu.:  67569
##  Median : 130367   Median :2018-12-30   Median :130.0   Median : 135182
##  Mean   : 135530   Mean   :2018-12-30   Mean   :135.1   Mean   : 135130
##  3rd Qu.: 203083   3rd Qu.:2019-03-31   3rd Qu.:203.0   3rd Qu.: 202652
##  Max.   :2373711   Max.   :2019-06-30   Max.   :272.0   Max.   :2415841
##     PROD_NBR       PROD_NAME            PROD_QTY       TOT_SALES
##  Min.   :  1.00   Length:246740      Min.   :1.000   Min.   : 1.700
##  1st Qu.: 26.00   Class :character   1st Qu.:2.000   1st Qu.: 5.800
##  Median : 53.00   Mode  :character   Median :2.000   Median : 7.400
##  Mean   : 56.35                      Mean   :1.906   Mean   : 7.316
##  3rd Qu.: 87.00                      3rd Qu.:2.000   3rd Qu.: 8.800
##  Max.   :114.00                      Max.   :5.000   Max.   :29.500
##    PACK_SIZE        BRAND            LIFESTAGE         PREMIUM_CUSTOMER
```

```
##  Min.   : 70.0   Length:246740      Length:246740      Length:246740
##  1st Qu.:150.0   Class :character   Class :character   Class :character
##  Median :170.0   Mode  :character   Mode  :character   Mode  :character
##  Mean   :175.6
##  3rd Qu.:175.0
##  Max.   :380.0
```

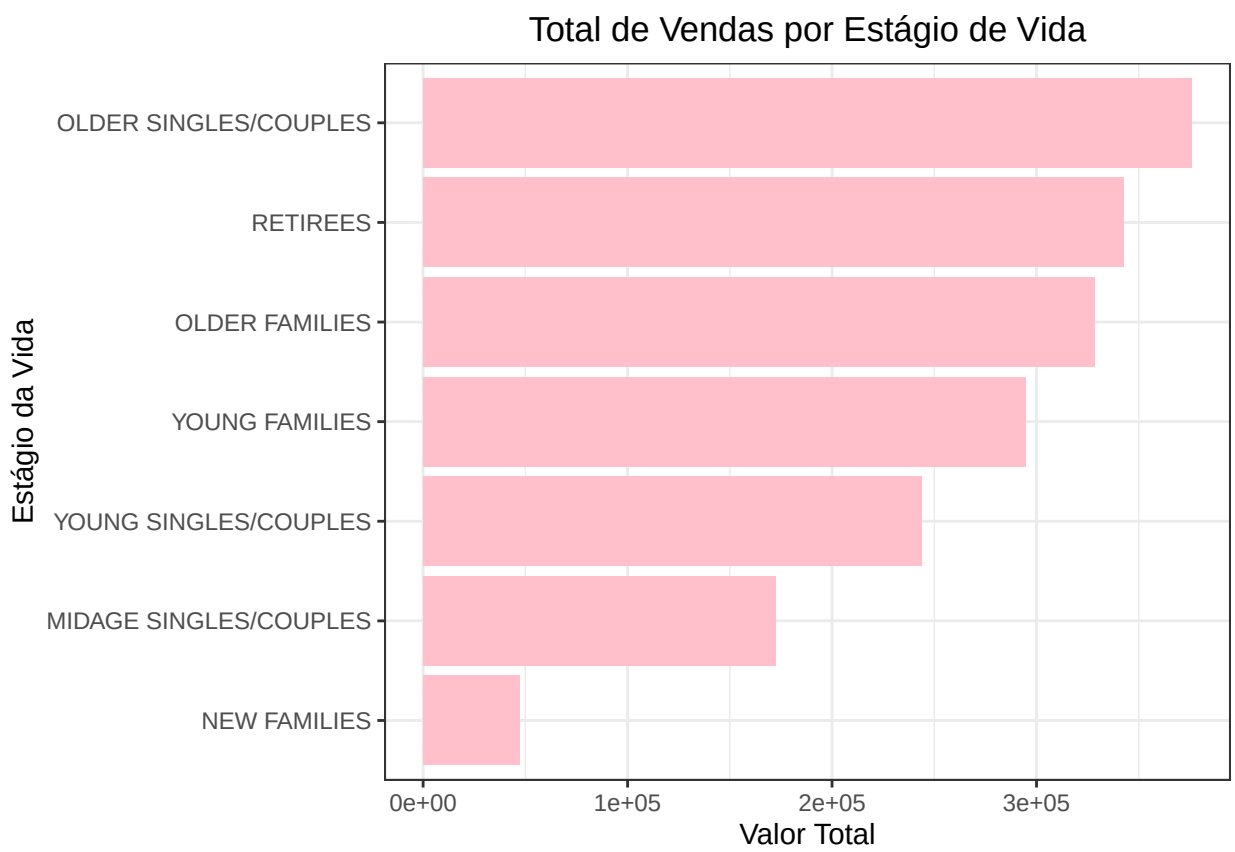
Salvar o dataset data em csv

```
caminho <- file.path("c:","Users", "renat", "Documents", "quantium", "quantium_chips")
fwrite(data, paste0(caminho, "QVI_data.csv"))
```

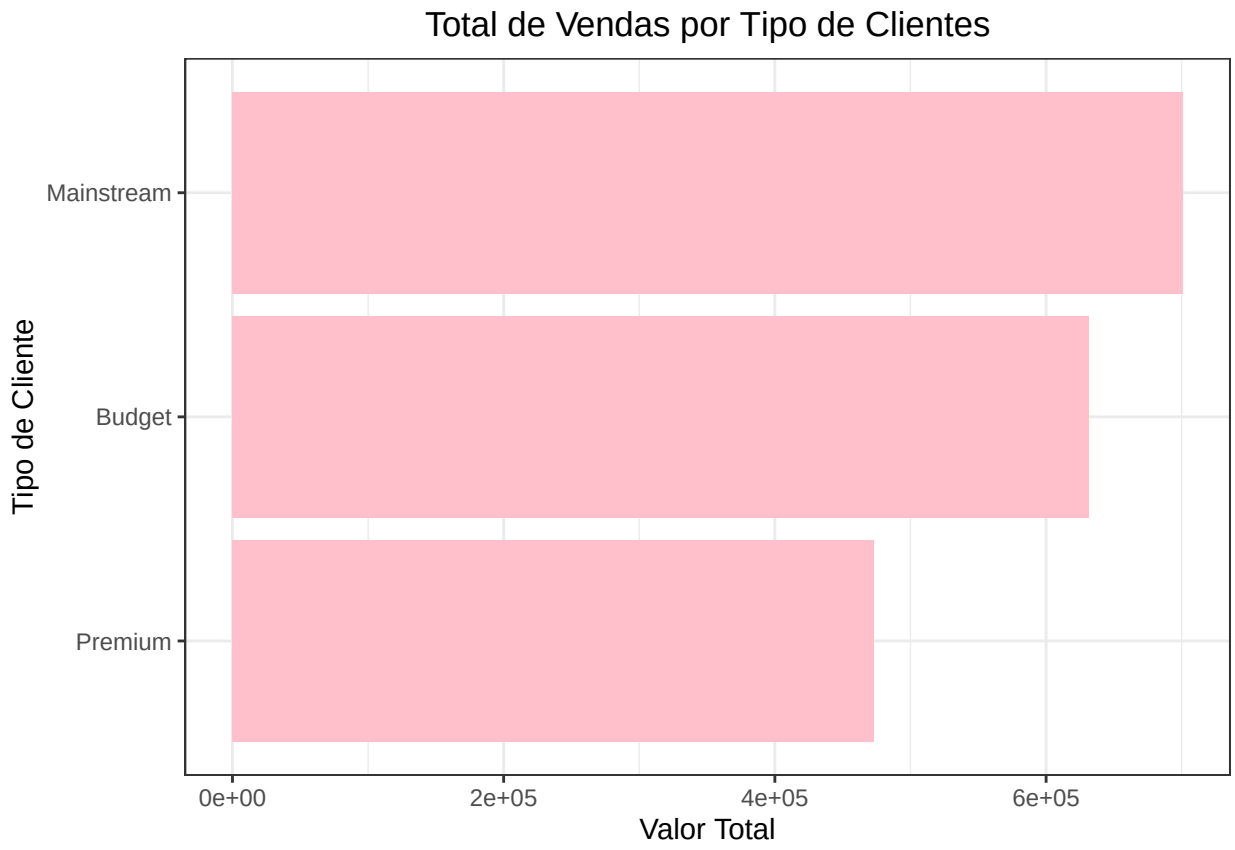
Parte de analise de dados

Total de vendas por estágio de vida(LIFE_STAGE) e clientes Premium(PREMIUM_CUSTOMER)

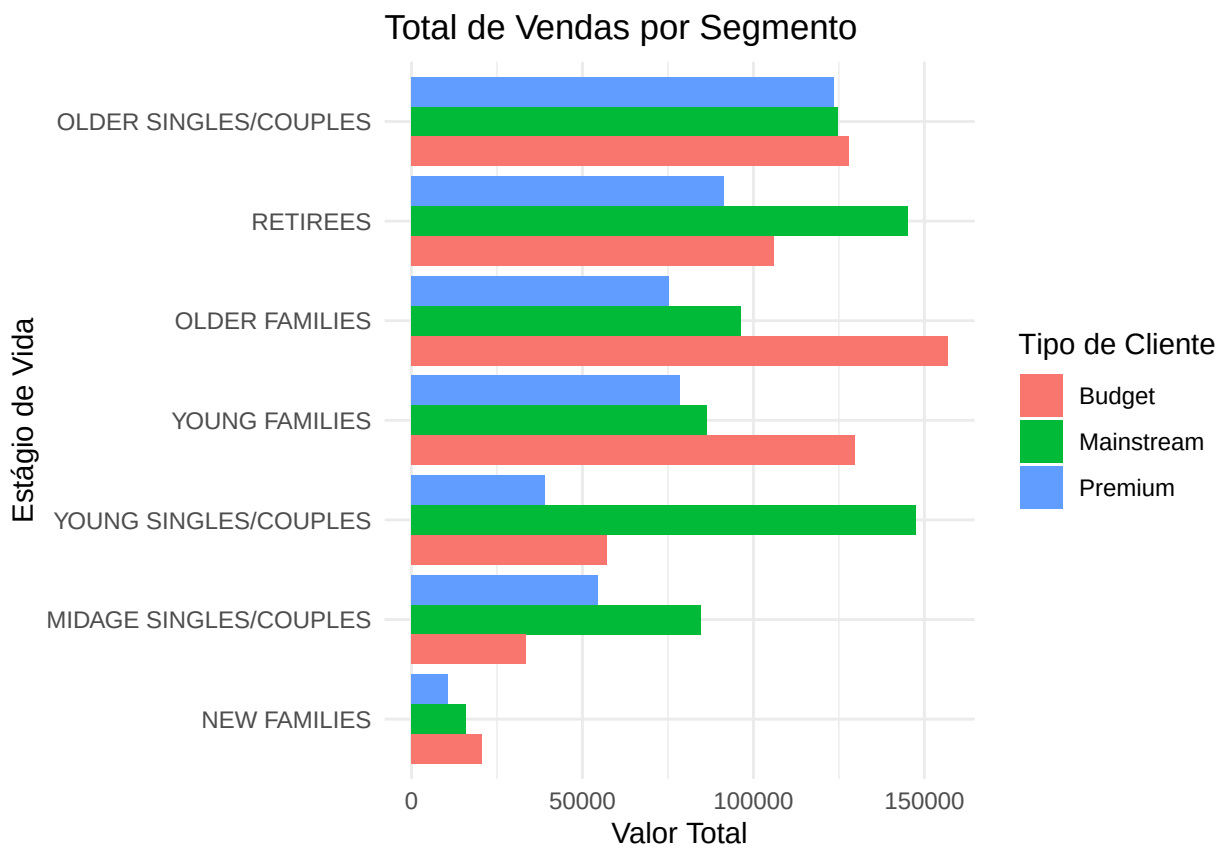
```
totVendasLife <- data[,.(Valor = sum(TOT_SALES)),by = LIFESTAGE]
totVendasPrem <- data[,.(Valor = sum(TOT_SALES)),by = PREMIUM_CUSTOMER]
totVendasLiPr <- data[,.(Valor = sum(TOT_SALES)),by = .(LIFESTAGE, PREMIUM_CUSTOMER)]


ggplot(totVendasLife, aes(x = Valor, y = reorder(LIFESTAGE, Valor))) +
  geom_bar(stat = "identity", fill = "pink") +
  labs(title = "Total de Vendas por Estágio de Vida" ,x = "Valor Total", y = "Estágio da Vida")
```



```r
ggplot(totVendasPrem, aes(x = Valor, y = reorder(PREMIUM_CUSTOMER, Valor))) +
  geom_bar(stat = "identity", fill = "pink" ) +
  labs(title = "Total de Vendas por Tipo de Clientes", x = "Valor Total", y = "Tipo de Cliente")
```

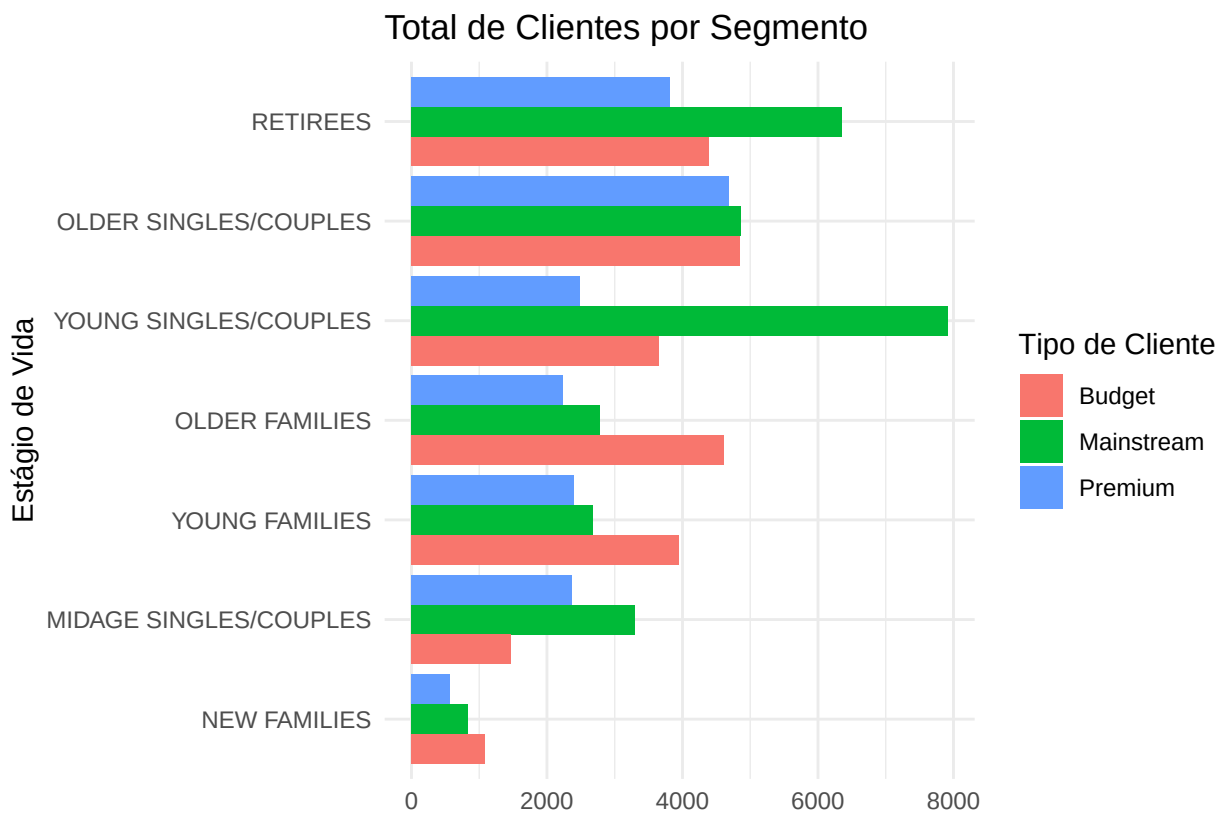



```r
ggplot(totVendasLiPr, aes(x = Valor, y = reorder(LIFESTAGE, Valor), fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Total de Vendas por Segmento",
      x = "Valor Total", y = "Estágio de Vida", fill = "Tipo de Cliente") +
  theme_minimal()
```



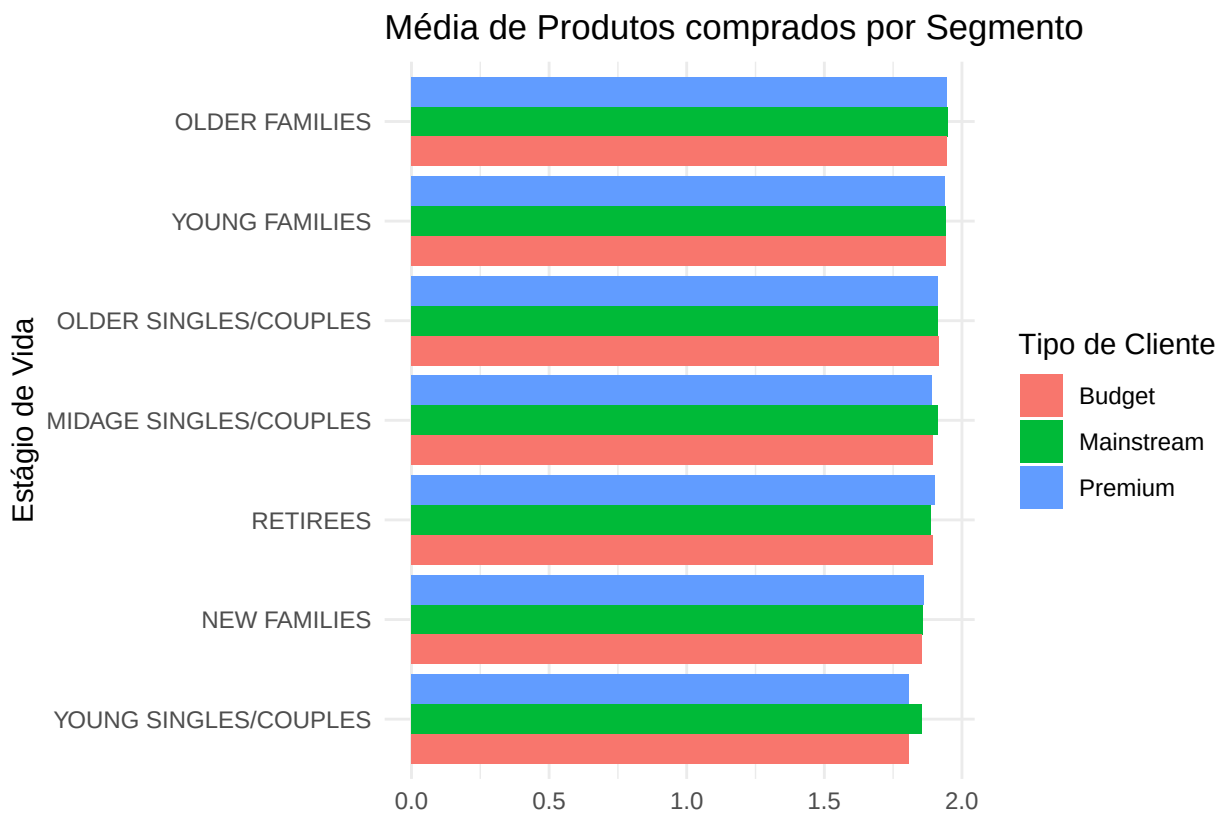
Número de clientes por Lifestage e Premium_Customer

```
LifePremN <- data[, .(N = uniqueN(LYLTY_CARD_NBR)), by = .(LIFESTAGE, PREMIUM_CUSTOMER)]

ggplot(LifePremN, aes(x = N, y = reorder(LIFESTAGE, N), fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Total de Clientes por Segmento",
       x = " ", y = "Estágio de Vida", fill = "Tipo de Cliente") +
  theme_minimal()
```



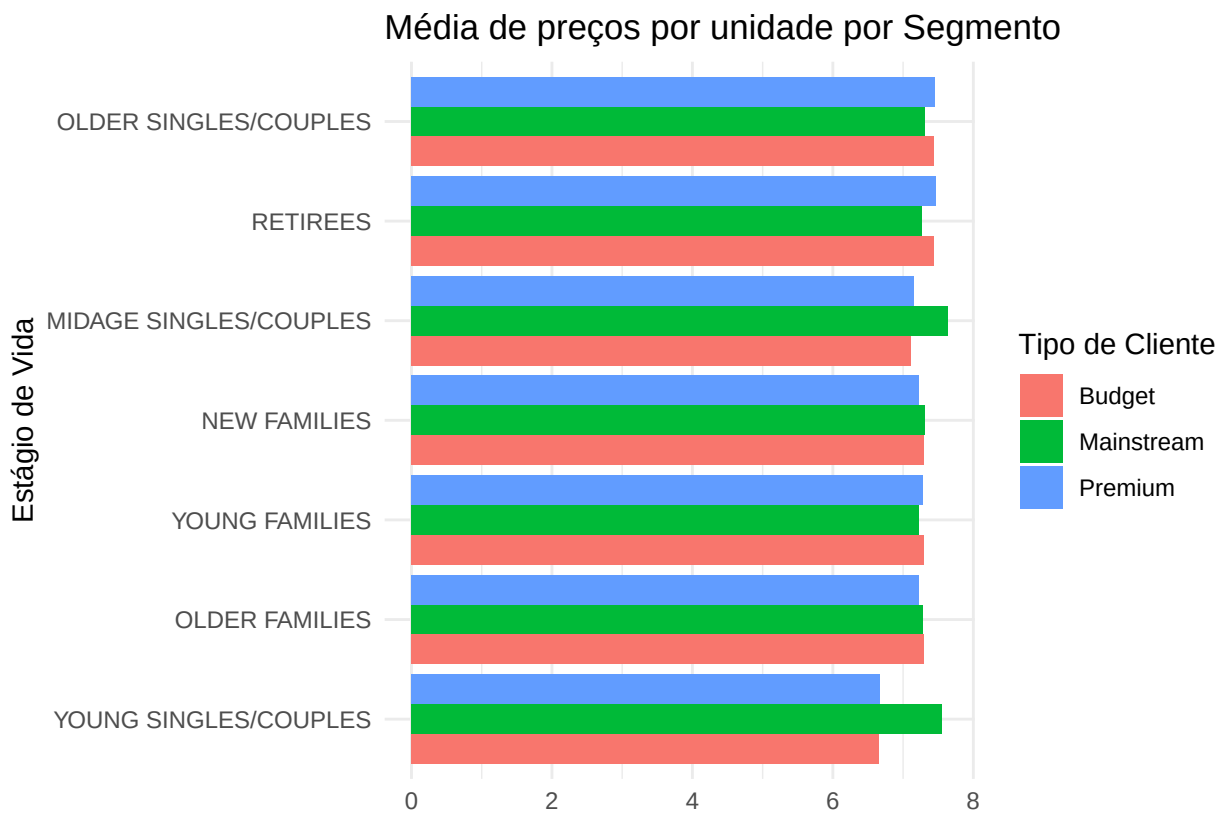
Média de número de produtos comprados por clientes, por LIFESTAGE e PREMIUM_CUSTOMER

```
LiPrMean <- data[, .(Media = mean(PROD_QTY)), by = .(LIFESTAGE, PREMIUM_CUSTOMER)]

ggplot(LiPrMean, aes(x = Media, y = reorder(LIFESTAGE, Media), fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Média de Produtos comprados por Segmento",
      x = " ", y = "Estágio de Vida", fill = "Tipo de Cliente") +
  theme_minimal()
```



Média dos preços por unidade de chips comprado por cliente, por LIFESTAGE e PREMIUM_CUSTOMER

```
priceMean <- data[, .(Media = mean(TOT_SALES)), by = .(LIFESTAGE, PREMIUM_CUSTOMER)]

ggplot(priceMean, aes(x = Media, y = reorder(LIFESTAGE, Media), fill = PREMIUM_CUSTOMER)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Média de preços por unidade por Segmento",
       x = " ", y = "Estágio de Vida", fill = "Tipo de Cliente") +
  theme_minimal()
```



Vamos checar se a diferença de preço por unidade é estatisticamente diferente dos grupos Mainstream versus Budget e Premium, de Young Singles/Couples e Midage Singles/Couples , utilizando o teste t.

```
#YoungBxM
t.test(data$TOT_SALES ~ data$PREMIUM_CUSTOMER,
                   data = data,
                   subset = (data$LIFESTAGE == "YOUNG SINGLES/COUPLES" & data$PREMIUM_CUSTOMER %in% c("
```

```
##
##  Welch Two Sample t-test
##
## data:  data$TOT_SALES by data$PREMIUM_CUSTOMER
## t = -26.536, df = 15428, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Budget and group Mainstream is not eq
## 95 percent confidence interval:
##  -0.9538673 -0.8226442
## sample estimates:
##     mean in group Budget mean in group Mainstream
##                 6.663023                 7.551279
```

```
#YoungPxM
t.test(data$TOT_SALES ~ data$PREMIUM_CUSTOMER,
                   data = data,
                   subset = (data$LIFESTAGE == "YOUNG SINGLES/COUPLES" & data$PREMIUM_CUSTOMER %in% c("
```

```
##
```

```
##  Welch Two Sample t-test
##
## data:  data$TOT_SALES by data$PREMIUM_CUSTOMER
## t = 22.63, df = 9112, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Mainstream and group Premium is not e
## 95 percent confidence interval:
##  0.8019057 0.9540019
## sample estimates:
## mean in group Mainstream    mean in group Premium
##                 7.551279                 6.673325
```

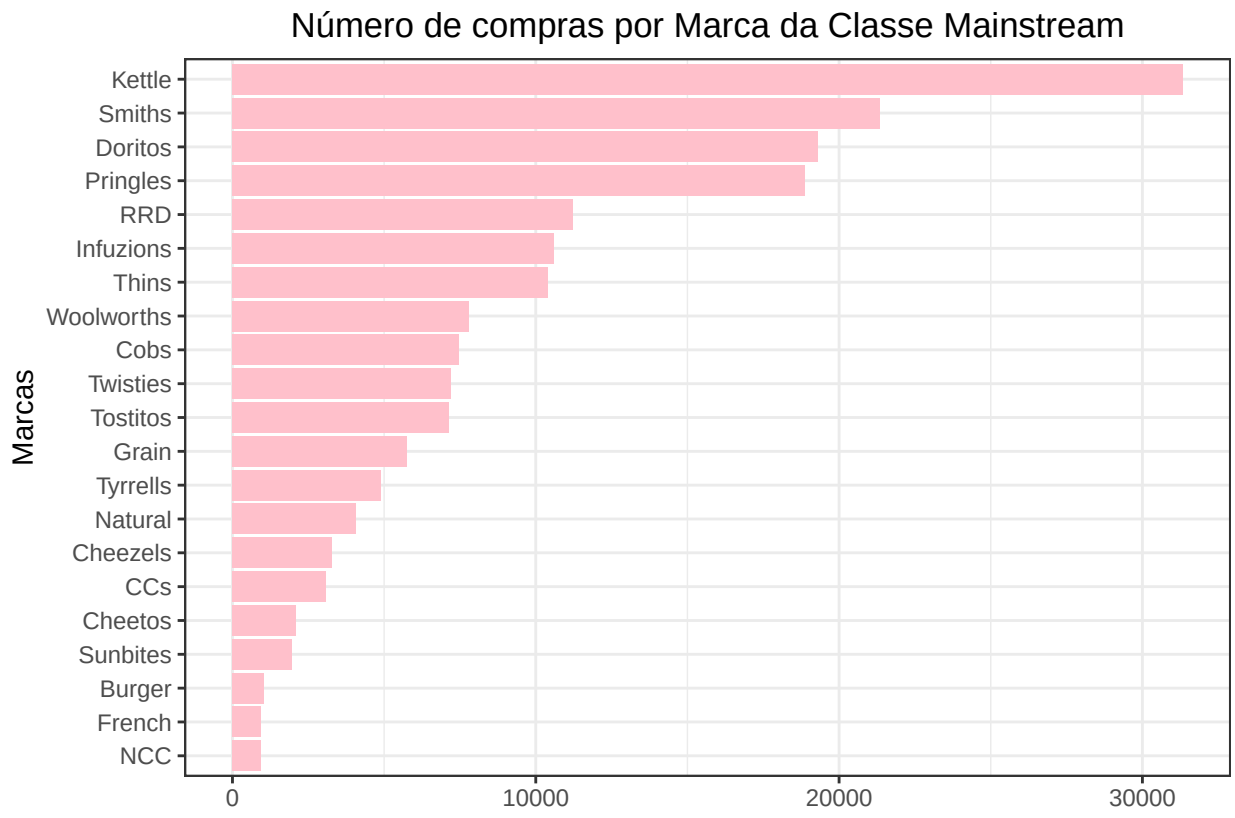
```r
#MidBxM
t.test(data$TOT_SALES ~ data$PREMIUM_CUSTOMER,
                   data = data,
                   subset = (data$LIFESTAGE == "MIDAGE SINGLES/COUPLES" & data$PREMIUM_CUSTOMER %in% c(
```

```
##
##  Welch Two Sample t-test
##
## data:  data$TOT_SALES by data$PREMIUM_CUSTOMER
## t = -12.213, df = 8504.6, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Budget and group Mainstream is not eq
## 95 percent confidence interval:
##  -0.6135734 -0.4438559
## sample estimates:
##     mean in group Budget mean in group Mainstream
##                 7.108442                 7.637156
```

```r
#MidPxM
t.test(data$TOT_SALES ~ data$PREMIUM_CUSTOMER,
                   data = data,
                   subset = (data$LIFESTAGE == "MIDAGE SINGLES/COUPLES" & data$PREMIUM_CUSTOMER %in% c(
```

```
##
##  Welch Two Sample t-test
##
## data:  data$TOT_SALES by data$PREMIUM_CUSTOMER
## t = 13.129, df = 15858, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Mainstream and group Premium is not e
## 95 percent confidence interval:
##  0.4124087 0.5571616
## sample estimates:
## mean in group Mainstream    mean in group Premium
##                 7.637156                 7.152371
```

Para um nível de significância igual à 0,05 , rejeitamos a hipótese que as diferença das médias é zero para todos os casos, ou seja, as médias são estatisticamente diferentes em todos os casos.

Vamos focar em segmentos de clientes que mais contribuem nas vendas, para retê-los ou aumentar ainda mais as vendas. Analizaremos o Mainstream e Young Single/Couples. Vamos ver se eles tendem a comprar um tipo de marca específica.

```r
PremBrands <- data[, .(N = sum(PROD_QTY)), by = .(PREMIUM_CUSTOMER , BRAND)]
MainBrands <- PremBrands[PREMIUM_CUSTOMER == "Mainstream"]

ggplot(MainBrands, aes(x = N, y = reorder(BRAND, N))) +
  geom_bar(stat = "identity", fill = "pink") +
  labs(title = "Número de compras por Marca da Classe Mainstream" , x = " ", y = "Marcas")
```